

4.1 Project Objectives

The primary objectives of the AgriTrace platform are:

1. **Develop a Role-Based Platform:** Create a web application with distinct interfaces and workflows for Farmers, Collection Agents, and Administrators.
2. **Implement Waste Tracking:** Enable end-to-end tracking of agricultural waste from reporting through collection, transit, and recycling.
3. **Build a Marketplace:** Provide a listing system where farmers can offer waste and agents can claim collection tasks.
4. **Integrate Payment Processing:** Implement Razorpay-based payment processing for waste transactions.
5. **Carbon Credit Tracking:** Calculate and track carbon offsets generated by waste diversion using scientifically established emission factors.
6. **Analytics Dashboard:** Provide comprehensive analytics including waste-by-type distribution, collection-over-time trends, and platform statistics.
7. **Location-Based Services:** Implement GPS-based location tracking for waste reports and collection coordination.
8. **AI Integration:** Leverage Google Genkit AI for intelligent waste classification and recommendations.
9. **Gamification:** Implement a rewards system with points, milestones, and streaks to incentivize platform participation.

4.2 Scope of the Project

The scope of AgriTrace encompasses:

- **User Management:** Registration, authentication, role selection, profile management, and password reset functionality.
- **Farmer Module:** Waste reporting with photo uploads, listing creation, order tracking, carbon credit viewing, and payment receipt.
- **Agent Module:** Task discovery via nearby-tasks feature, listing claiming, pickup-to-delivery workflow management, delivery proof uploads, and earnings tracking.
- **Admin Module:** User management (edit roles, view all users), platform analytics, listing oversight, recycling status management, and statistics dashboard.